



In this chapter we set out from a type of problem which, on the face of it, appears to be similar to the theme of Chapter 7: what kind of substructures are necessarily present in every large enough graph?

While ‘large’ in Chapter 7 meant ‘many edges’ on a given number of vertices, in this chapter it will simply mean ‘many vertices’. Of course, we cannot expect any concrete graph H to arise as a subgraph in just any graph G of large enough order. But if it does not, maybe we can say something else about G ? For example, if $H \not\subseteq G$ then maybe G contains some other graph we can specify in advance, one very different from H ?

Ramsey’s theorem, in its simplest form, says just that: for every integer $r \geq 0$, every large enough graph G contains either a K^r or $\overline{K}^r$ as an induced subgraph. If G is connected, we can say a little more: then G contains either a K^r or a large induced path or star. What can we say if G is 2-connected, or even 3-connected?

Despite its superficial similarity to extremal problems, the above type of question leads to a kind of mathematics with a distinctive flavour of its own. Indeed, the theorems and proofs in this chapter have more in common with similar results in algebra or geometry, say, than with most other areas of graph theory. The study of their underlying methods, therefore, is generally regarded as a combinatorial subject in its own right: the discipline of *Ramsey theory*.

In line with the subject of this book, we shall focus on results that are naturally expressed in terms of graphs. Even from the viewpoint of general Ramsey theory, however, this is not as much of a limitation as it might seem: graphs are a natural setting for Ramsey problems, and the material in this chapter brings out a sufficient variety of ideas and methods to convey some of the fascination of the theory as a whole.

9.1 Ramsey's original theorems

In its simplest version, Ramsey's theorem says that, given an integer $r \geq 0$, every large enough graph G contains either a K^r or a $\overline{K}^r$ as an induced subgraph. At first glance, this may seem surprising: after all, we need a proportion of about $(r-2)/(r-1)$ of all possible edges to force a K^r subgraph in G (Corollary 7.1.3), but neither G nor $\overline{G}$ can be expected to have more than half of all possible edges.

However, there is a subtle difference: we are not trying to force either of these two events, that $G \supseteq K^r$ or that $\overline{G} \supseteq K^r$, but only that one or the other happens. Keeping an open mind as to which of them does, rather looking for a K^r only in the denser half of the set of potential edges, helps G to live up to our expectations. And Ramsey's theorem says that it only has to be large enough.

Turán's theorem fits this pattern too when G has exactly $t_{r-1}(n)$ edges on n vertices. In this case it tells us more than Ramsey's theorem: if G does not contain a K^r then $\overline{G}$ contains a $K^{\lceil n/(r-1) \rceil}$, a complete graph of order much larger than r .

So how could we go about proving Ramsey's theorem? Let us try to build a K^r or $\overline{K}^r$ in G inductively, starting with an arbitrary vertex $v_1 \in V_1 := V(G)$. If $|G|$ is large, there will be a large set $V_2 \subseteq V_1 \setminus \{v_1\}$ of vertices that are either all adjacent to v_1 or all non-adjacent to v_1 . Accordingly, we may think of v_1 as the first vertex of a K^r or $\overline{K}^r$ whose other vertices all lie in V_2 . Let us then choose another vertex $v_2 \in V_2$ for our K^r or $\overline{K}^r$. Since V_2 is large, it will have a subset V_3 , still fairly large, of vertices that are all 'of the same type' with respect to v_2 as well: either all adjacent or all non-adjacent to it. We then continue our search for vertices inside V_3 , and so on (Fig. 9.1.1).

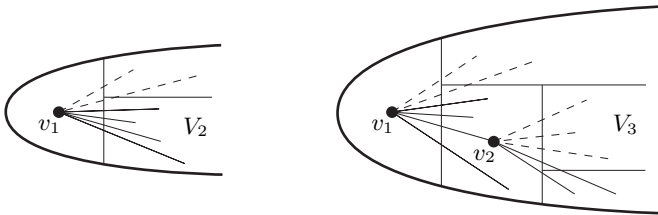


Fig. 9.1.1. Choosing the sequence $v_1, v_2, \dots$

How long can we go on in this way? This depends on the size of our initial set V_1 : each set V_i has at least half the size of its predecessor V_{i-1} , so we shall be able to complete s construction steps if G has order about 2^s . As the following proof shows, the choice of $s = 2r - 3$ vertices v_i suffices to find among them the vertices of a K^r or $\overline{K}^r$.

Theorem 9.1.1. (Ramsey 1930)

[9.2.2]

For every $r \in \mathbb{N}$ there exists an $n \in \mathbb{N}$ such that every graph of order at least n contains either K^r or $\overline{K}^r$ as an induced subgraph.

Proof. The assertion is trivial for $r \leq 1$; we assume that $r \geq 2$. Let $n := 2^{2r-3}$, and let G be a graph of order at least n . We shall define a sequence $V_1, \dots, V_{2r-2}$ of sets and choose vertices $v_i \in V_i$ with the following properties:

- (i) $|V_i| = 2^{2r-2-i}$ ($i = 1, \dots, 2r-2$);
- (ii) $V_i \subseteq V_{i-1} \setminus \{v_{i-1}\}$ ($i = 2, \dots, 2r-2$);
- (iii) v_{i-1} is adjacent either to all vertices in V_i or to no vertex in V_i ($i = 2, \dots, 2r-2$).

Let $V_1 \subseteq V(G)$ be any set of 2^{2r-3} vertices, and pick $v_1 \in V_1$ arbitrarily. Then (i) holds for $i = 1$, while (ii) and (iii) hold trivially. Suppose now that V_{i-1} and $v_{i-1} \in V_{i-1}$ have been chosen so as to satisfy (i)–(iii) for $i-1$, where $1 < i \leq 2r-2$. Since

$$|V_{i-1} \setminus \{v_{i-1}\}| = 2^{2r-1-i} - 1$$

is odd, V_{i-1} has a subset V_i satisfying (i)–(iii); we pick $v_i \in V_i$ arbitrarily.

Among the $2r-3$ vertices $v_1, \dots, v_{2r-3}$, there are $r-1$ vertices that show the same behaviour when viewed as v_{i-1} in (iii), being adjacent either to all the vertices in V_i or to none. Accordingly, these $r-1$ vertices and v_{2r-2} induce either a K^r or a $\overline{K}^r$ in G , because $v_i, \dots, v_{2r-2} \in V_i$ for all i . $\square$

The least integer n associated with r as in Theorem 9.1.1 is the *Ramsey number* $R(r)$ of r . Our proof shows that $R(r) \leq 2^{2r-3}$, which is close to the best known upper bound; see the Notes. In Chapter 11 we shall use a simple probabilistic argument to show that $R(r)$ is bounded below by $2^{r/2}$ (Theorem 11.1.3).

Ramsey number
 $R(r)$

In other words, the clique and independence number of a graph can both, asymptotically, be logarithmically small compared with its order. As soon as we forbid some fixed induced subgraph, however, one of them may have to be bigger: The *Erdős-Hajnal conjecture* says that for every graph H there exists a constant $\delta_H > 0$ such that every graph G not containing an induced copy of H has a set of at least $|G|^{\delta_H}$ vertices that are either independent or span a complete subgraph in G .

Erdős-Hajnal conjecture

It is customary in Ramsey theory to think of partitions as colourings: a *colouring* of (the elements of) a set X with c colours, or c -*colouring* for short, is simply a partition of X into c classes¹ (indexed by the ‘colours’). In particular, these colourings need not satisfy any non-adjacency

c-colouring

¹ Unlike elsewhere in this book, these partition classes may be empty.

$[X]^k$ mono-
chromatic

requirements as in Chapter 5. Given a c -colouring of $[X]^k$, the set of all k -subsets of X , we call a set $Y \subseteq X$ *monochromatic* if all the elements of $[Y]^k$ have the same colour,² i.e. belong to the same of the c partition classes of $[X]^k$. Similarly, if $G = (V, E)$ is a graph and all the edges of $H \subseteq G$ have the same colour in some colouring of E , we call H a *monochromatic subgraph* of G , speak of a red (green, etc.) H in G , and so on.

In the above terminology, Ramsey's theorem can be expressed as follows: for every r there exists an n such that, given any n -set X , every 2-colouring of $[X]^2$ yields a monochromatic r -set $Y \subseteq X$. Interestingly, this assertion remains true for c -colourings of $[X]^k$ with arbitrary c and k – with almost exactly the same proof!

We first prove the infinite version, which is easier, and then deduce the finite version.

[12.1.1]

Theorem 9.1.2. *Let k, c be positive integers, and X an infinite set. If $[X]^k$ is coloured with c colours, then X has an infinite monochromatic subset.*

Proof. We prove the theorem by induction on k , with c fixed. For $k = 1$ the assertion holds, so let $k > 1$ and assume the assertion for smaller values of k .

Let $[X]^k$ be coloured with c colours. We shall construct an infinite sequence $X_0, X_1, \dots$ of infinite subsets of X and choose elements $x_i \in X_i$ with the following properties (for all i):

- (i) $X_{i+1} \subseteq X_i \setminus \{x_i\}$;
- (ii) all k -sets $\{x_i\} \cup Z$ with $Z \in [X_{i+1}]^{k-1}$ have the same colour, which we *associate* with x_i .

We start with $X_0 := X$ and pick $x_0 \in X_0$ arbitrarily. By assumption, X_0 is infinite. Having chosen an infinite set X_i and $x_i \in X_i$ for some i , we c -colour $[X_i \setminus \{x_i\}]^{k-1}$ by giving each set Z the colour of $\{x_i\} \cup Z$ from our c -colouring of $[X]^k$. By the induction hypothesis, $X_i \setminus \{x_i\}$ has an infinite monochromatic subset, which we choose as X_{i+1} . Clearly, this choice satisfies (i) and (ii). Finally, we pick $x_{i+1} \in X_{i+1}$ arbitrarily.

Since c is finite, one of the c colours is associated with infinitely many x_i . These x_i form an infinite monochromatic subset of X . $\square$

If desired, the finite version of Theorem 9.1.2 could be proved just like the infinite version above. However to ensure that the relevant sets are large enough at all stages of the induction, we have to keep track of their sizes, which involves a good deal of boring calculation. As long as we are not interested in bounds, the more elegant route is to deduce the finite version from the infinite ‘by compactness’, that is, using König’s infinity lemma (8.1.2).

² Note that Y is called monochromatic, but it is the elements of $[Y]^k$, not of Y , that are (equally) coloured.

Theorem 9.1.3. *For all $k, c, r \geq 1$ there exists an $n \geq k$ such that every n -set X has a monochromatic r -subset with respect to any c -colouring of $[X]^k$.* [9.3.3]

Proof. As is customary in set theory, we denote by $n \in \mathbb{N}$ (also) the set $\{0, \dots, n-1\}$. Suppose the assertion fails for some k, c, r . Then for every $n \geq k$ there exist an n -set, without loss of generality the set n , and a c -colouring $[n]^k \rightarrow c$ such that n contains no monochromatic r -set. Let us call such colourings *bad*; we are thus assuming that for every $n \geq k$ there exists a bad colouring of $[n]^k$. Our aim is to combine these into a bad colouring of $[\mathbb{N}]^k$, which will contradict Theorem 9.1.2. (8.1.2)
 k, c, r

For every $n \geq k$ let $V_n \neq \emptyset$ be the set of bad colourings of $[n]^k$. For $n > k$, the restriction $f(g)$ of any $g \in V_n$ to $[n-1]^k$ is still bad, and hence lies in V_{n-1} . By the infinity lemma (8.1.2), there is an infinite sequence $g_k, g_{k+1}, \dots$ of bad colourings $g_n \in V_n$ such that $f(g_n) = g_{n-1}$ for all $n > k$. For every $m \geq k$, all colourings g_n with $n \geq m$ agree on $[m]^k$, so for each $Y \in [\mathbb{N}]^k$ the value of $g_n(Y)$ coincides for all $n > \max Y$. Let us define $g(Y)$ as this common value $g_n(Y)$. Then g is a bad colouring of $[\mathbb{N}]^k$: every r -set $S \subseteq \mathbb{N}$ is contained in some sufficiently large n , so S cannot be monochromatic since g coincides on $[n]^k$ with the bad colouring g_n . $\square$

The least integer n associated with k, c, r as in Theorem 9.1.3 is the *Ramsey number* for these parameters; we denote it by $R(k, c, r)$. Ramsey number
 $R(k, c, r)$

9.2 Ramsey numbers

Ramsey's theorem may be rephrased as follows: if $H = K^r$ and G is a graph with sufficiently many vertices, then either G itself or its complement $\overline{G}$ contains a copy of H as a subgraph. Clearly, the same is true for any graph H , simply because $H \subseteq K^h$ for $h := |H|$.

However, if we ask for the *least* n such that every graph G of order n has the above property – this is the *Ramsey number* $R(H)$ of H – then the above question makes sense: if H has only few edges, it should embed more easily in G or $\overline{G}$, and we would expect $R(H)$ to be smaller than the Ramsey number $R(h) = R(K^h)$. Ramsey number
 $R(H)$

A little more generally, let $R(H_1, H_2)$ denote the least $n \in \mathbb{N}$ such that $H_1 \subseteq G$ or $H_2 \subseteq \overline{G}$ for every graph G of order n . For most graphs H_1, H_2 , only very rough estimates are known for $R(H_1, H_2)$. Interestingly, lower bounds given by random graphs (as in Theorem 11.1.3) are often sharper than even the best bounds provided by explicit constructions. $R(H_1, H_2)$

The following proposition describes one of the few cases where exact Ramsey numbers are known for a relatively large class of graphs: